Cyclic Sort

When given numbers from range 1 to N → Use cyclic sort.

Q) What is cyclic sort and how does it work? Eg) 3, 5, 2, 1, 4

3, 5, 2, 1, 4 → Here, the nos. are jumbled, but are from 1 to 5.

lets say, N = 5

When the array is sorted, all numbers are going to be at their correct indices.

After sorting:

1,	2,	3,	4,	5
0	1	2	3	4

{ Here, after sorting, the index will become -1 }

Index = Value - 1 {Using this we will sort the array}

↳ Because index starts from 0.

→ WORKING

Eg)

3,	5,	2,	1,	4
0	1	2	3	4

① Check if 3 is at the correct index and if not, 3 - 1 = 2 (index = value - 1) Then swap with correct index.

2, 5, 3, 1, 4

→ After swapping we know that 3 is at correct position now but we do not know whether the other number that came at the position of 3 is correct or not. So check again.

② If 2 is at correct position, in this case it is not. Swap it again.

2, 5, 3, 1, 4

5, 2, 3, 1, 4

4, 2, 3, 1, 5

$\boxed{1, 2, 3, 4, 5}$

→ We know that every unique item is only getting swapped once.

→ Here, we are not incrementing 'i' when we are swapping so that might result in more than 'n' iterations of the loop.

→ Worst case: (N-1) + N

→ O(n) [Linear]